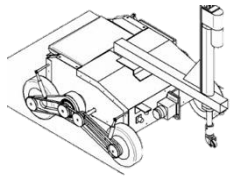
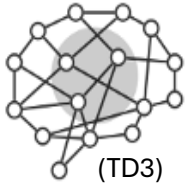
Curriculum

Seguimiento y Curriculum



Agente1900

(...) →  (TD3)

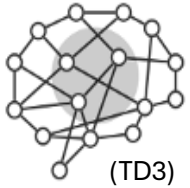
Scen58, Reward_SystemA
Run_x860, 2000eps
Checkpoint1900

↓

Entrenamiento:
Scen58, Reward_SystemA
Run_x519, 1000eps

Agente 500

Agente seleccionado
Checkpoint500

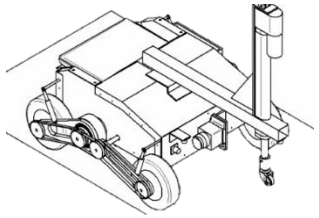


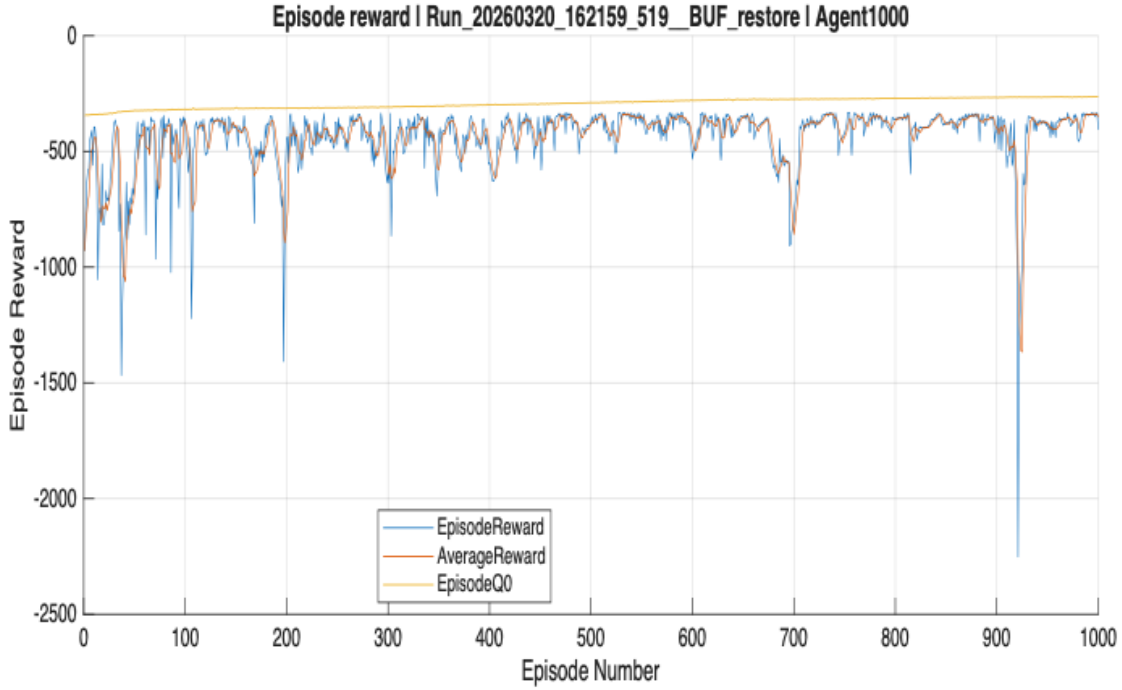
(TD3)

Evaluación del Aprendizaje

Universidad Internacional de Valencia

Curva de Aprendizaje y Métricas. Agente1000





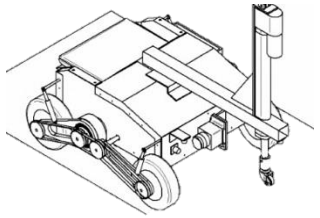
Training Stats			Advanced Agent Options		
Métrica	Símbolo	Valor	Métrica	Símbolo	Valor
Total de episodios	N_{ep}	1000	Tiempo muestreo	T_s	0.01
Recompensa total último episodio	R_{ep}	-406.147	Factor descuento	γ	0.99
Recompensa Media	$\bar{R}$	-352.339	Tamaño del Batchize	B	64
Pasos último episodio	-	600	Tamaño del Exp. Buffer		10000
Número total de pasos	N_{steps}	600000	Modo del Buffer	-	Reset
Promedio pasos/eps	-	600	Taza aprendizaje actor	α_a	1.0000e-04
Valor Q estimado	Q_0	-263.797	Optimizador del actor	-	adam
			Taza aprendizaje critico	α_c	0.001
			Optimizador del critico	-	adam
			Coeficiente actualización suave	τ	0.005

Exploration Gaussian Model			Target Policy Smoothing		
Métrica	Símbolo	Valor	Métrica	Símbolo	Valor
Std.Dev	σ	0.3162	Std.Dev	σ	0.2
Mean	μ	0	Mean	μ	0
Lower Lim	-	-0.5	Lower Lim	-	-0.5
Upper Lim	-	0.5	Upper Lim	-	0.5
Std.Dev decay	λ_σ	1.9187e-06	Std.Dev decay	λ_σ	0
Std.Dev min	σ_{min}	0.1	Std.Dev min	σ_{min}	0.2

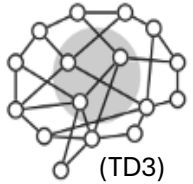
Evaluación del Aprendizaje

Universidad Internacional de Valencia

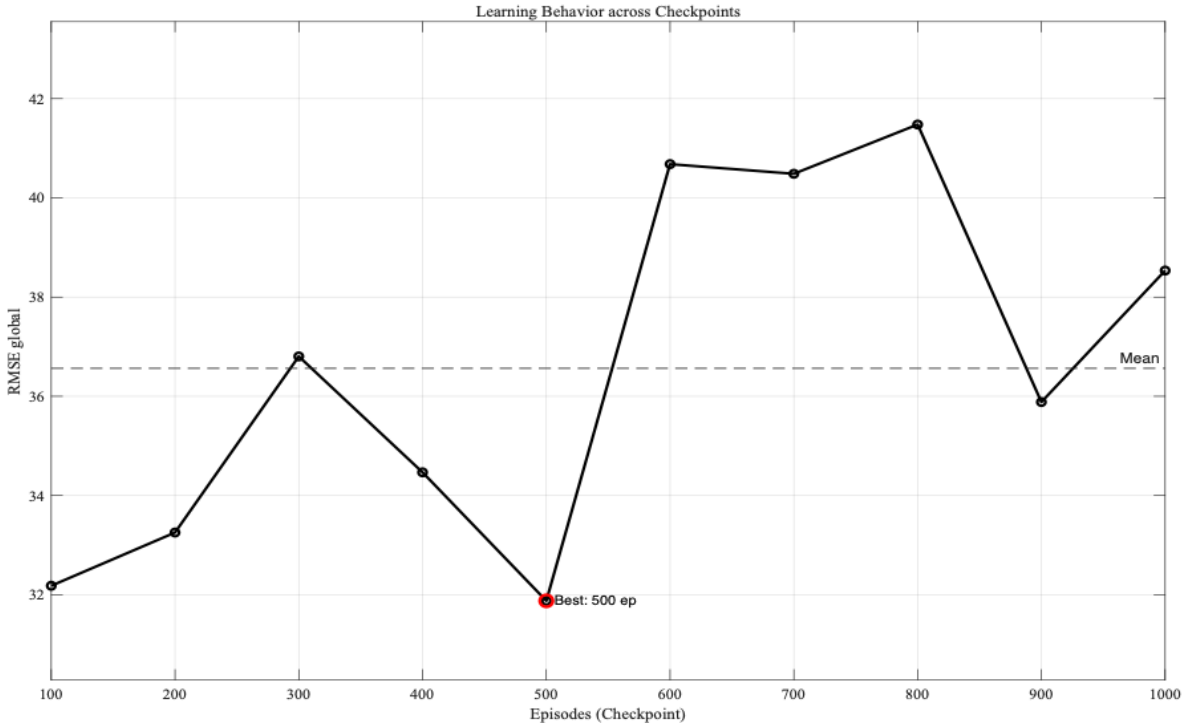
Progresión del RMSE durante el entrenamiento.



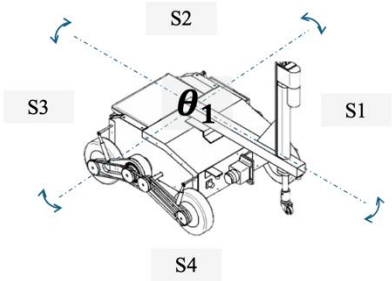
Agentes RL



(TD3)



Efecto Policy Degration, Overfitting



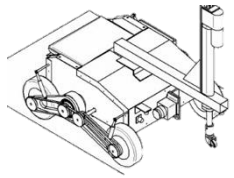
Estado baseline: Pitch = 0°, Roll = 0° , o1ini=0° , do1ini=0rad/s

El mejor punto de aprendizaje se alcanza en 500 episodios; a partir de ahí, el rendimiento empeora debido a sobreajuste.

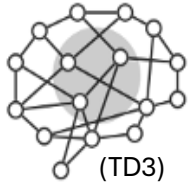
Evaluación del Aprendizaje

Universidad Internacional de Valencia

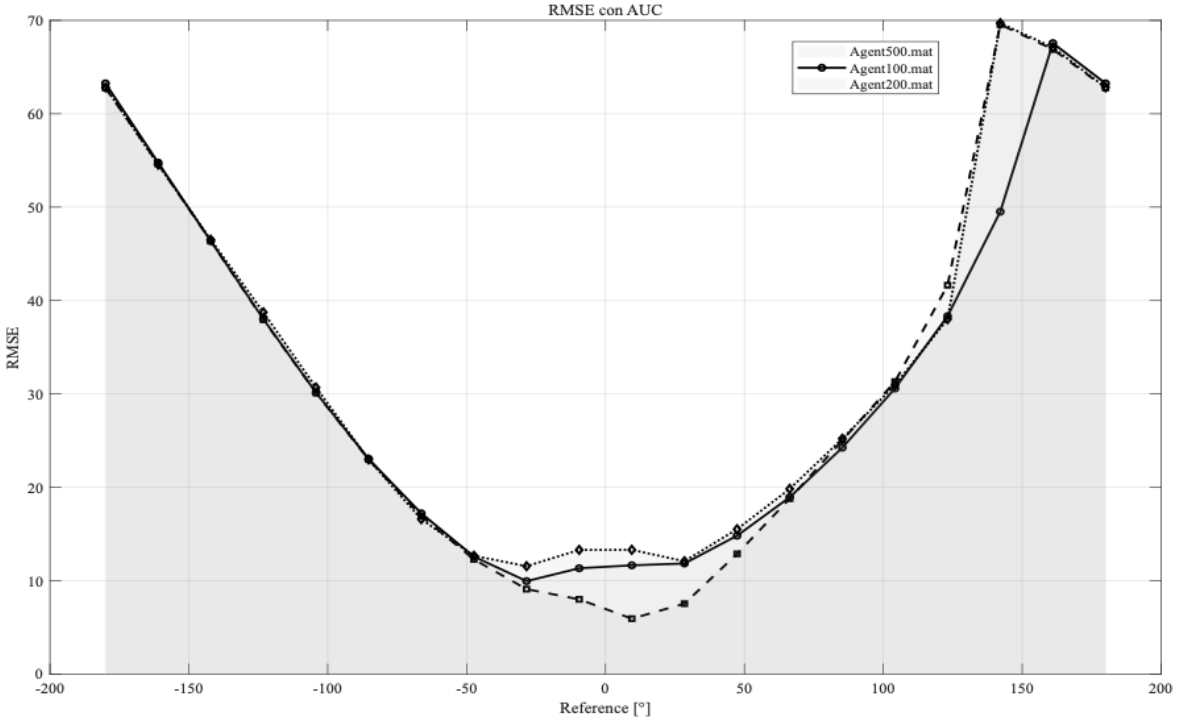
Top3 Curvas RMSE en el Dominio Angular



Agentes RL



(TD3)

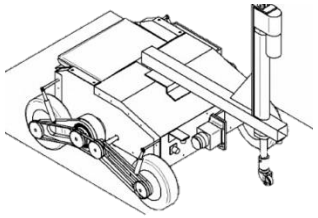


El RMSE presenta una distribución suave y consistente en todo el dominio angular, con mínimos en la zona central y aumento progresivo hacia los extremos, evidenciando una política estable y generalizada.

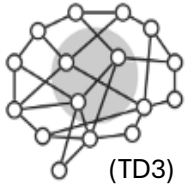
Evaluación del Aprendizaje

Universidad Internacional de Valencia

Implementación de **RMSE geodésico**



Entrenamiento:
Scen58, Reward_SystemA
Run_x519, 1000eps



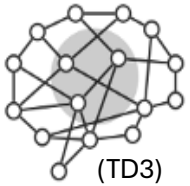
Agente seleccionado
Checkpoint500

Tabla con posición de los agentes según sus métricas

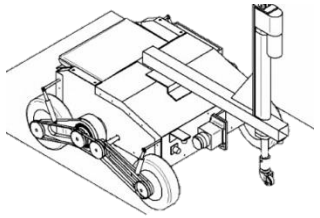
Position	Agente	$RMSE_{global}$	$RMSE_{max}$	$\sigma_{angular}$	$Ratio_{ext}$	AUC
1	Agent500.mat	31.8768	67.5612	19.5546	2.1194	30.2252
2	Agent100.mat	32.1803	69.5560	21.8054	2.1614	30.5703
3	Agent200.mat	33.2539	69.7003	20.5150	2.0960	31.6975
4	Agent400.mat	34.4669	69.6161	19.0805	2.0198	32.9844
5	Agent900.mat	35.8842	69.5453	17.6581	1.9380	34.5013
6	Agent300.mat	36.8039	69.6267	17.0374	1.8918	35.4449
7	Agent1000.mat	38.5345	69.7105	16.2425	1.8090	37.2744
8	Agent700.mat	40.4853	75.5817	17.5543	1.8669	39.3504
9	Agent600.mat	40.6800	71.7127	17.5065	1.7629	39.5563
10	Agent800.mat	41.4784	71.6549	17.5960	1.7275	40.3722

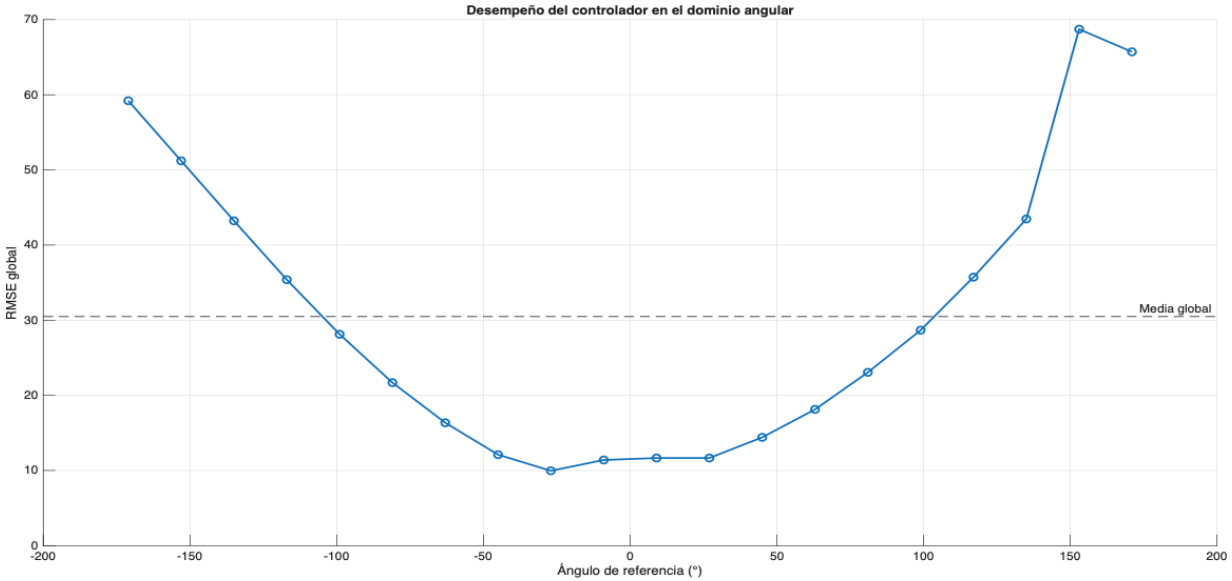
Los Agentes evaluados corresponden a diferentes Checkpoints del entrenamiento del agente RL. El Análisis de múltiples checkpoints permite observar la mejora progresiva del agente durante el entrenamiento. **En este caso, el Agente500 mantiene la mejor posición en el ranking, al presentar los menores valores en las métricas de error y mejor compromiso global de desempeño.**

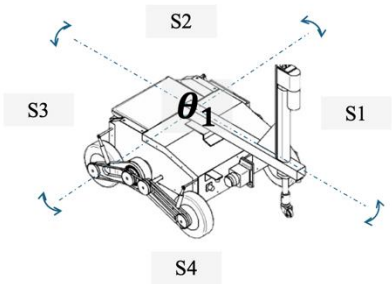
Evaluación Determinista en la Simulación



$RMSE_i$ en dominio angular.







Estado baseline: Pitch = 0°, Roll = 0° , o1ini=0° , do1ini=0rad/s

- El agente500 presenta un comportamiento suave y continuo en todo el dominio angular, sin discontinuidades abruptas como en el experimento anterior El seguimiento es preciso y estable, indicando correcta adaptación de la política.
- El error aumenta progresivamente hacia los extremos del dominio.
- No se observan saltos abruptos, pero sí una degradación sistemática del rendimiento.

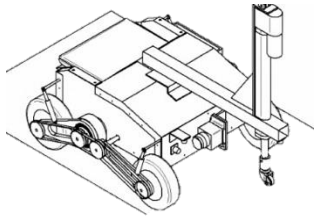
El comportamiento es aproximadamente simétrico respecto al eje central, lo que indica que el agente ha aprendido una política más consistente que en etapas anteriores.

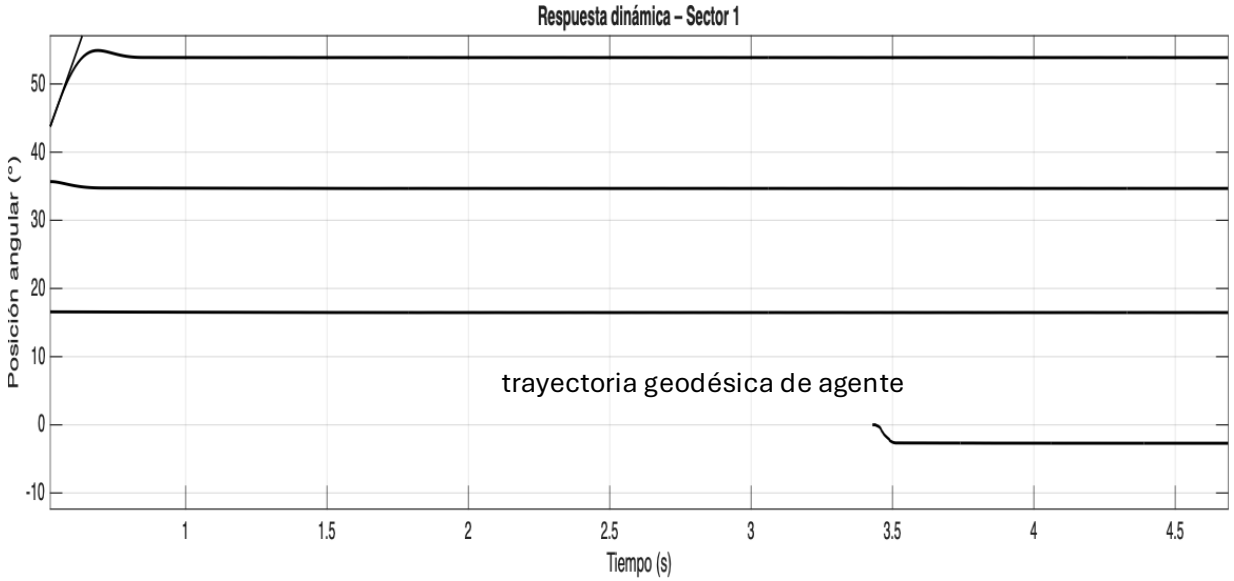
Evaluación Determinista en la Simulación

Universidad Internacional de Valencia

(TD3)

Respuesta dinámica del sector 1

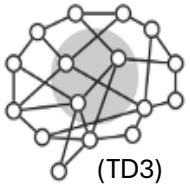




Estado baseline: Pitch = 0°, Roll = 0° , o1ini=0° , do1ini=0rad/s

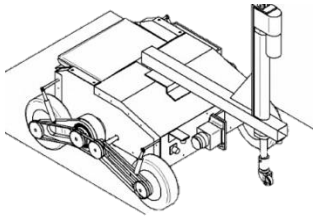
Rank	O1sp	O1_final	Mp [°]	Mp[%]	T[s]	e_ss_abs[°]	RMSE[°]	ITAE[°s]	Tr[s]	Iosc	U_peak	U_rms	U_energy	U_abs	U_TV
1	9.00	-2.71	-9.00	-100.00	6.00	11.71	11.66	141.38	NaN	0.02	9.45	0.52	116.34	56.06	16.62
2	27.00	16.47	-10.02	-37.13	6.00	10.53	11.66	189.77	NaN	0.13	9.86	1.78	1849.38	262.53	47.70
3	45.00	34.67	-9.33	-20.72	6.00	10.33	14.41	187.31	NaN	0.23	9.99	2.50	3720.50	448.89	49.66
4	63.00	53.87	-8.09	-12.84	6.00	9.13	18.13	168.46	NaN	0.30	10.00	3.00	5365.68	612.64	51.40
5	81.00	74.12	-5.52	-6.81	6.00	6.88	23.06	132.76	0.59	0.36	10.00	3.49	7284.07	800.71	54.60

Evaluación Determinista en la Simulación

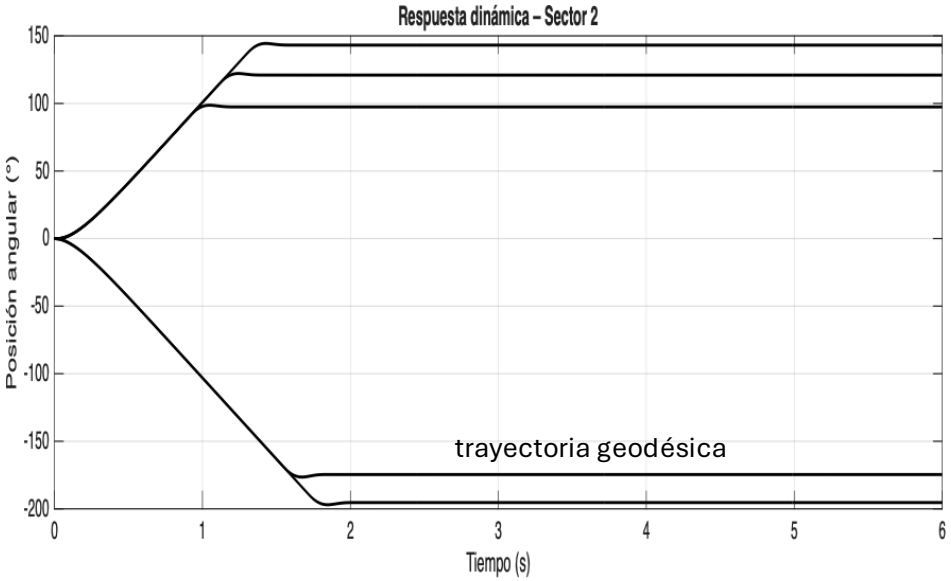


(TD3)

Respuesta dinámica del sector 2



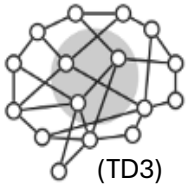
La trayectoria observada no representa un error de control, sino una solución equivalente en el dominio angular circular, donde el agente sigue **una trayectoria geodésica** hacia la referencia.



Estado baseline: Pitch = 0°, Roll = 0° , o1ini=0° , do1ini=0rad/s

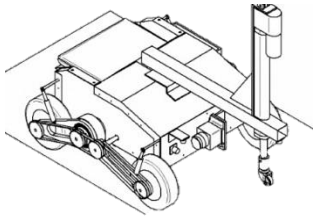
Rank	O1sp	O1_final	Mp [°]	Mp[%]	T[s]	e_ss_abs[°]	RMSE[°]	ITAE[°s]	Tr[s]	Iosc	U_peak	U_rms	U_energy	U_abs	U_TV
1	99.00	97.41	-0.32	-0.32	0.98	1.59	28.67	46.48	0.70	0.42	10.00	3.94	9264.02	996.41	53.39
2	117.00	120.97	5.16	4.41	6.00	3.97	35.73	97.89	0.81	0.47	10.00	4.34	11266.88	1195.66	54.52
3	135.00	143.15	9.40	6.96	6.00	8.15	43.47	181.80	0.93	0.51	10.00	4.63	12842.83	1366.92	54.83
4	171.00	-174.61	-171.00	-100.00	6.00	14.38	65.71	339.46	NaN	0.57	10.00	5.15	15885.89	1653.71	36.43
5	153.00	-195.36	-153.00	-100.00	6.00	11.64	68.73	318.07	NaN	0.61	10.00	5.38	17348.54	1805.66	35.14

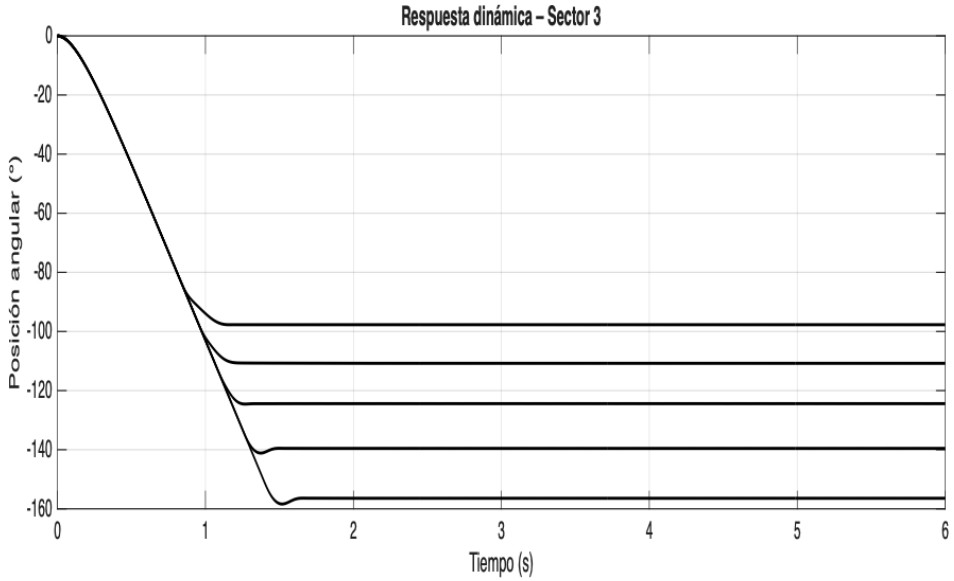
Evaluación Determinista en la Simulación



(TD3)

Respuesta dinámica del sector 3





Estado baseline: Pitch = 0°, Roll = 0° , o1ini=0° , do1ini=0rad/s

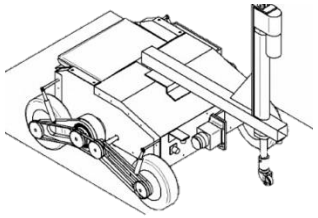
Rank	O1sp	O1_final	Mp [°]	Mp[%]	T[s]	e_ss_abs[°]	RMSE[°]	ITAE[°s]	Tr[s]	Iosc	U_peak	U_rms	U_energy	U_abs	U_TV
1	-99.00	-97.71	99.00	100.00	1.08	1.29	28.13	40.95	0.00	0.40	10.00	3.72	8281.02	896.96	44.82
2	-117.00	-110.77	117.00	100.00	6.00	6.23	35.39	136.39	0.00	0.44	10.00	3.96	9383.71	1020.65	40.48
3	-135.00	-124.41	135.00	100.00	6.00	10.59	43.23	223.11	0.00	0.47	10.00	4.25	10788.43	1148.69	39.95
4	-153.00	-139.55	153.00	100.00	6.00	13.44	51.20	285.86	0.00	0.51	10.00	4.64	12885.84	1356.16	44.45
5	-171.00	-156.43	171.00	100.00	6.00	14.57	59.21	321.92	0.00	0.54	10.00	4.92	14500.28	1514.55	40.96

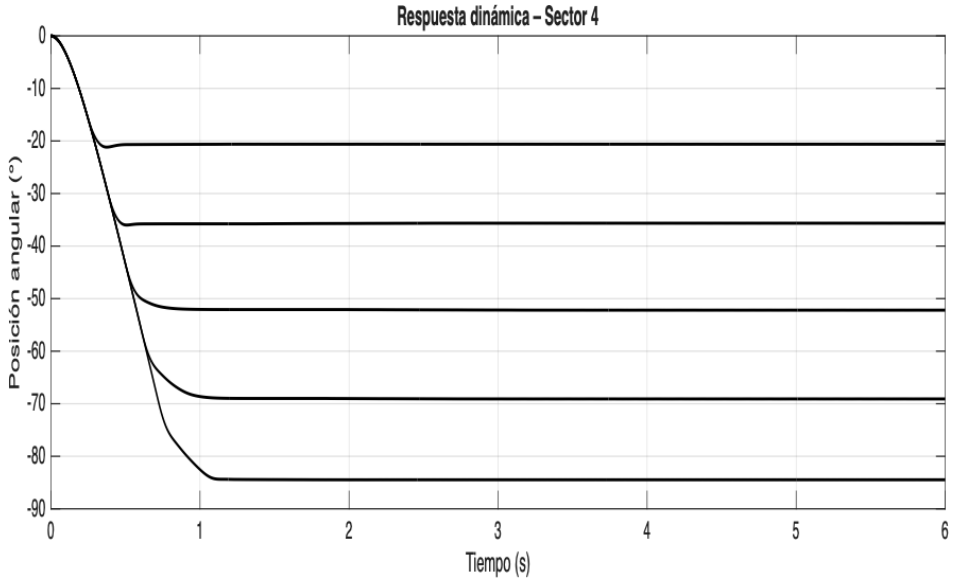
Evaluación Determinista en la Simulación

Universidad Internacional de Valencia

(TD3)

Respuesta dinámica del sector 4





Estado baseline: Pitch = 0°, Roll = 0° , o1ini=0° , do1ini=0rad/s

Rank	O1sp	O1_final	Mp [°]	Mp[%]	T[s]	e_ss_abs[°]	RMSE[°]	ITAE[°s]	Tr[s]	Iosc	U_peak	U_rms	U_energy	U_abs	U_TV
1	-27.00	-35.65	27.00	100.00	6.00	8.65	9.96	155.90	0.00	0.23	10.00	2.58	3952.54	481.99	35.68
2	-9.00	-20.62	9.00	100.00	6.00	11.61	11.40	208.82	0.00	0.16	10.00	2.03	2422.99	309.84	32.45
3	-45.00	-52.20	45.00	100.00	6.00	7.19	12.10	130.30	0.00	0.28	10.00	2.91	5035.10	599.87	36.45
4	-63.00	-69.08	63.00	100.00	6.00	6.08	16.35	112.86	0.00	0.33	10.00	3.20	6103.75	694.11	38.30
5	-81.00	-84.47	81.00	100.00	6.00	3.47	21.69	71.52	0.00	0.37	10.00	3.47	7204.23	788.90	50.35